

Tong Statistical Field Theory — Sheet 3: Renormalisation Group

D. Tong, Statistical Field Theory, Example Sheet 3; 6 worked problems.

Yucheng (Luca) Jin

Thermodynamics & Statistical Physics

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These prompts paraphrase David Tong’s Statistical Field Theory, Example Sheet 3 (November 2018). Problem numbers, starred status, equations, and guidance follow the official sheet.

1. Sheet 3, Problem 1: Critical Exponents of the $O(N)$ Model

For N real scalars in $d = 4 - \epsilon$, take

$$F_1[\phi] = \int d^d x \left[\frac{1}{2} (\nabla \phi)^2 + \frac{\mu^2}{2} (\phi \cdot \phi) + g (\phi \cdot \phi)^2 \right]. \quad (1.1)$$

The supplied flows for $\tilde{g} = \Lambda^{-\epsilon} g$ are

$$\frac{d\mu^2}{ds} = 2\mu^2 + \frac{N+2}{2\pi^2} \frac{\Lambda^4}{\Lambda^2 + \mu^2} \tilde{g}, \quad \frac{d\tilde{g}}{ds} = \epsilon \tilde{g} - \frac{N+8}{2\pi^2} \frac{\Lambda^4}{(\Lambda^2 + \mu^2)^2} \tilde{g}^2. \quad (1.2)$$

Find ν at the Wilson–Fisher fixed point. Assuming $\eta = O(\epsilon^2)$, determine $\alpha, \beta, \gamma, \delta$ to leading order in ϵ .

Solution. To first order in ϵ , the Wilson–Fisher fixed point is given by

$$\mu_*^2 = -\frac{1}{2} \frac{N+2}{N+8} \epsilon \Lambda^2, \quad \tilde{g}_* = \frac{2\pi^2}{N+8} \epsilon. \quad (1.3)$$

Let us consider the RG flow near the W–F fixed point, which is

$$\frac{d}{ds} \begin{pmatrix} \delta\mu^2 \\ \delta\tilde{g} \end{pmatrix} = \begin{pmatrix} 2 - \frac{N+2}{N+8} \epsilon & \frac{N+2}{2\pi^2} \left(1 + \frac{1}{2} \frac{N+2}{N+8} \epsilon\right) \Lambda^2 \\ 0 & -\epsilon \end{pmatrix} \begin{pmatrix} \delta\mu^2 \\ \delta\tilde{g} \end{pmatrix}, \quad (1.4)$$

which has eigenvalues $\lambda_1 = 2 - \frac{N+2}{N+8} \epsilon$ and $\lambda_2 = -\epsilon$. Since $\delta\mu^2 \sim t$, the first eigenvalue is precisely $\Delta_t = \lambda_1$. Then, by

$$\xi \rightarrow \zeta^{-1} \xi \sim \zeta^{-1} t^{-\nu}, \quad t \rightarrow \zeta^{\Delta_t} t, \quad (1.5)$$

we find

$$\nu = \frac{1}{\Delta_t} = \frac{1}{2} + \frac{1}{4} \frac{N+2}{N+8} \epsilon + O(\epsilon^2). \quad (1.6)$$

From this, we can evaluate the remaining critical exponents.

$$c \sim \frac{\partial^2 f}{\partial t^2}, \quad (1.7)$$

while $F = \int d^d f$, and hence $[f(t)] = -d$. Therefore, $f \sim \xi^{-d} \sim t^{\nu d}$, which implies $c \sim t^{\nu d - 2}$, namely

$$\alpha = 2 - \nu d = \frac{4 - N}{2(N+8)} \epsilon + O(\epsilon^2). \quad (1.8)$$

Then, consider

$$\phi \sim \xi^{-\Delta_\phi} \sim t^{\nu \Delta_\phi} \implies \beta = \nu \Delta_\phi = \frac{1}{2} - \frac{3}{2(N+8)} \epsilon + O(\epsilon^2), \quad (1.9)$$

where we have used the fact that $\eta = \mathcal{O}(\epsilon^2)$. By the term $\int d^d x B \phi$, we now that $\Delta_\phi + \Delta_B = d$, and

$$\chi = \frac{\partial \phi}{\partial B} \sim \xi^{-\Delta_\phi + \Delta_B} \sim t^{\nu(\Delta_\phi - \Delta_B)}, \quad (1.10)$$

which implies

$$\gamma = -\nu(\Delta_\phi - \Delta_B) = -\nu(2\Delta_\phi - d) = 1 + \frac{N+2}{2(N+8)}\epsilon + \mathcal{O}(\epsilon^2). \quad (1.11)$$

Finally,

$$\phi \sim \xi^{-\Delta_\phi} \sim B^{\Delta_\phi/\Delta_B} \implies \delta = \frac{\Delta_B}{\Delta_\phi} = 3 + \epsilon + \mathcal{O}(\epsilon^2). \quad (1.12)$$

2. Sheet 3, Problem 2: Flow Between Gaussian and Wilson–Fisher Fixed Points

Along the flow of Problem 1, $\mu^2 = \mathcal{O}(\epsilon)$. At leading order, write

$$\frac{d\tilde{g}}{ds} \simeq \epsilon\tilde{g} - \frac{N+8}{2\pi^2}\tilde{g}^2 = -A\tilde{g}(\tilde{g} - g_\star). \quad (2.1)$$

Determine $A, g_\star$, then solve the flow from g_0 at $s = 0$ to obtain

$$\tilde{g}(s) = \frac{g_\star}{1 - (1 - g_\star/g_0)e^{-\epsilon s}}. \quad (2.2)$$

Solution. Clearly,

$$A = \frac{N+8}{2\pi^2}, \quad g_\star = \frac{2\pi^2}{N+8}\epsilon. \quad (2.3)$$

Then,

$$\int_{g_0}^{\tilde{g}} \frac{dg}{g(g - g_\star)} = -A \int_0^s ds' \implies \log \frac{1 - g_\star/\tilde{g}}{1 - g_\star/g_0} = -A g_\star s \implies \tilde{g}(s) = \frac{g_\star}{1 - (1 - g_\star/g_0)e^{-\epsilon s}}. \quad (2.4)$$

3. Sheet 3, Problem 3*: Cubic Anisotropy and Fixed-Point Stability

Add an $\mathcal{O}(N)$ -breaking interaction to Problem 1:

$$F_2[\phi] = F_1[\phi] + \int d^d x \lambda \sum_{a=1}^N \phi_a^4. \quad (3.1)$$

Establish the necessary positive definite condition $g + \lambda > 0$. For $\tilde{g} = \Lambda^{-\epsilon} g$ and $\tilde{\lambda} = \Lambda^{-\epsilon} \lambda$, use the given flows

$$\frac{d\tilde{g}}{ds} = \epsilon\tilde{g} - \frac{N+8}{2\pi^2}\tilde{g}^2 - \frac{3}{\pi^2}\tilde{g}\tilde{\lambda}, \quad \frac{d\tilde{\lambda}}{ds} = \epsilon\tilde{\lambda} - \frac{6}{\pi^2}\tilde{g}\tilde{\lambda} - \frac{9}{2\pi^2}\tilde{\lambda}^2. \quad (3.2)$$

Find the Gaussian, Heisenberg ($\tilde{\lambda} = 0$), Ising ($\tilde{g} = 0$), and cubic ($\tilde{g}\tilde{\lambda} \neq 0$) fixed points. Check the cubic point against the positive definite condition. Determine the stability of every point in the (g, λ) plane, and plot the RG trajectories separately for $N < 4$ and $N > 4$. *Hint:* The determinant and trace determine the eigenvalue signs of a 2×2 stability matrix.

Solution. For the free energy to be stable, we require the quartic part to be positive definite. Therefore, we must have

$$g \left(\sum_{a=1}^N \phi_a^2 \right)^2 + \lambda \sum_{a=1}^N \phi_a^4 > 0 \quad (3.3)$$

for all nonzero field configurations. Taking only one component to be nonzero implies $g + \lambda > 0$. By solving

$$\tilde{g} \left(\epsilon - \frac{N+8}{2\pi^2} \tilde{g} - \frac{3}{\pi^2} \tilde{\lambda} \right) = 0, \quad (3.4)$$

$$\tilde{\lambda} \left(\epsilon - \frac{6}{\pi^2} \tilde{g} - \frac{9}{2\pi^2} \tilde{\lambda} \right) = 0, \quad (3.5)$$

we find four fixed points:

$$\text{Gaussian : } \tilde{\lambda} = 0, \tilde{g} = 0, \quad (3.6)$$

$$\text{Heisenberg : } \tilde{\lambda} = 0, \tilde{g} = \frac{2\pi^2}{N+8} \epsilon, \quad (3.7)$$

$$\text{Ising : } \tilde{\lambda} = \frac{2\pi^2}{9} \epsilon, \tilde{g} = 0, \quad (3.8)$$

and

$$\text{Cubic : } \tilde{\lambda} = \frac{2\pi^2(N-4)}{9N} \epsilon, \tilde{g} = \frac{2\pi^2}{3N} \epsilon. \quad (3.9)$$

The cubic point satisfies this necessary condition for $N > 1$, since $\tilde{g} + \tilde{\lambda} = 2\pi^2(N-1)\epsilon/(9N) > 0$.

At GF, we have

$$\frac{d}{ds} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon \end{pmatrix} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix}, \quad (3.10)$$

so GF is unstable in both directions. At IF, we have

$$\frac{d}{ds} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} \epsilon/3 & 0 \\ -4\epsilon/3 & -\epsilon \end{pmatrix} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix}, \quad (3.11)$$

so it is unstable in the $(1, -1)$ direction and stable along the $\tilde{\lambda}$ axis. At HF, we have

$$\frac{d}{ds} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} -\epsilon & -6\epsilon/(N+8) \\ 0 & (N-4)\epsilon/(N+8) \end{pmatrix} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix}. \quad (3.12)$$

The eigenvalues are $-\epsilon$ and $(N-4)\epsilon/(N+8)$. Thus HF is stable for $N < 4$ and unstable in one direction for $N > 4$. Finally, at CF, we have

$$\frac{d}{ds} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix} = \begin{pmatrix} -(N+8)\epsilon/3N & -2\epsilon/N \\ -4(N-4)\epsilon/3N & -(N-4)\epsilon/N \end{pmatrix} \begin{pmatrix} \delta \tilde{g} \\ \delta \tilde{\lambda} \end{pmatrix}. \quad (3.13)$$

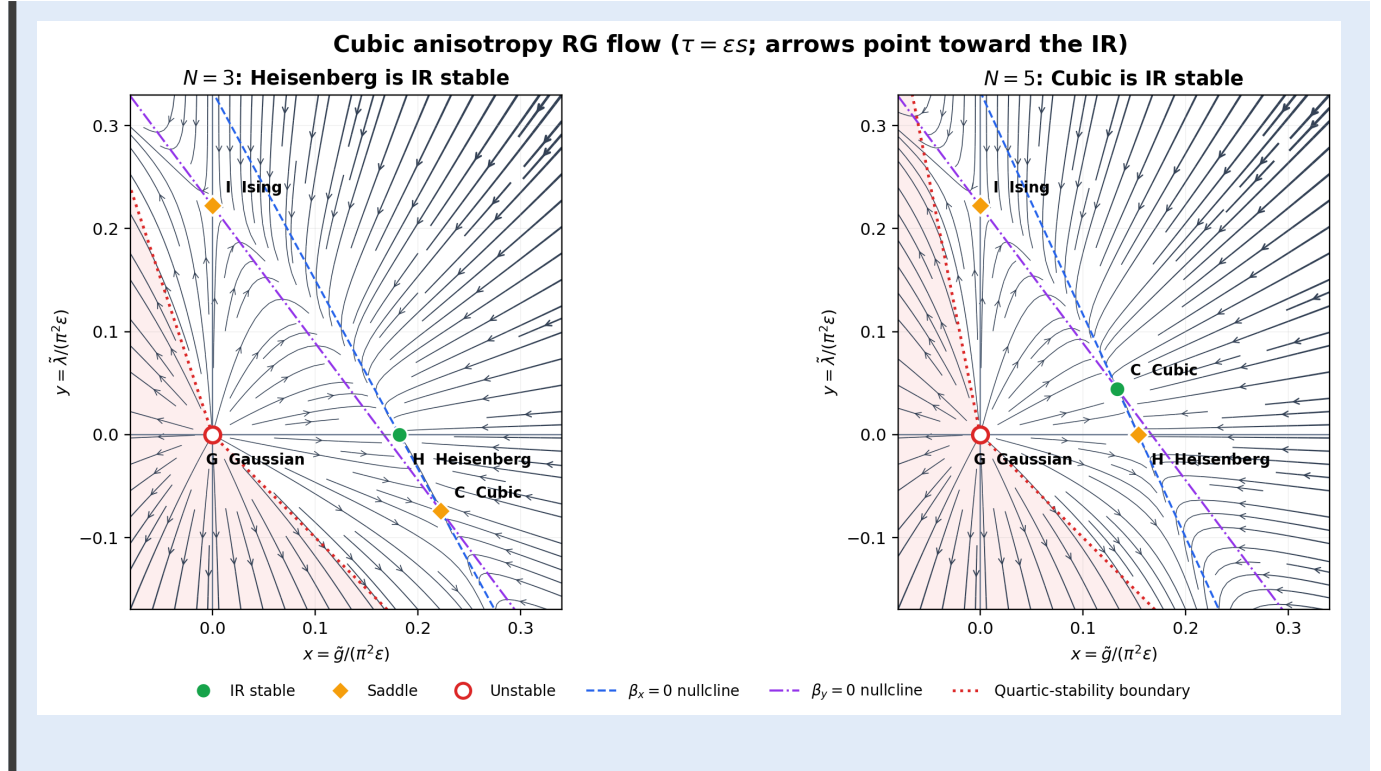
The matrix has eigenvectors and eigenvalues

$$\lambda_1 = -\epsilon, \quad v_1 = \begin{pmatrix} 3 \\ N-4 \end{pmatrix} \quad (3.14)$$

$$\lambda_2 = -\frac{N-4}{3N} \epsilon, \quad v_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}. \quad (3.15)$$

Therefore, for $N > 4$, CF is stable in all directions, while for $1 < N < 4$ it becomes unstable in the v_2 direction. At $N = 4$, HF and CF coincide and one eigenvalue vanishes.

From these we can draw the RG diagrams.



4. Sheet 3, Problem 4*: The Sine-Gordon Perturbation

In two dimensions with ultraviolet cutoff Λ , consider

$$F[\phi] = \int d^2x \left[\frac{1}{2} (\nabla \phi)^2 - \lambda_0 \cos(\beta_0 \phi) \right]. \quad (4.1)$$

Determine the engineering dimensions of β_0 and λ_0 . Split $\phi_{\mathbf{k}} = \phi_{\mathbf{k}}^- + \phi_{\mathbf{k}}^+$, with the plus modes supported on $\Lambda/\zeta < |k| < \Lambda$, and define their position-space fields. Integrate out the plus modes to first order in λ_0 to derive

$$F'[\phi^-] = \int d^2x \left[\frac{1}{2} (\nabla \phi^-)^2 - \lambda_0 \langle \cos(\beta_0(\phi^- + \phi^+)) \rangle_+ \right]. \quad (4.2)$$

Define $\langle \cdot \rangle_+$ and evaluate it, showing that the cosine coefficient becomes $\lambda_0 \zeta^{-\beta_0^2/(4\pi)}$. Hence find the relevance threshold $\beta_0^2 = 8\pi$. *Hint:* Expand the cosine into exponentials and use Wick's identity from Sheet 2, Problem 8, together with

$$\langle \phi^+(x) \phi^+(y) \rangle_+ = \int_{\Lambda/\zeta < |k| < \Lambda} \frac{d^2k}{(2\pi)^2} \frac{e^{-ik \cdot (x-y)}}{k^2} \quad (4.3)$$

in the coincident-point limit.

Solution. Naively, the engineering dimensions are $[\phi] = 0$, $[\beta_0] = 0$, and $[\lambda_0] = -2$. Splitting into $\phi_{\mathbf{k}} = \phi_{\mathbf{k}}^+ + \phi_{\mathbf{k}}^-$, the partition function becomes

$$\int_{k < \Lambda} \mathcal{D}\phi e^{-F[\phi]} = \int_{k < \Lambda/\zeta} \mathcal{D}\phi^- \int_{\Lambda/\zeta < k < \Lambda} \mathcal{D}\phi^+ e^{-F[\phi]} = \int_{k < \Lambda/\zeta} \mathcal{D}\phi^- e^{-F'[\phi^-]}, \quad (4.4)$$

where

$$e^{-F'[\phi^-]} = e^{-F_0[\phi^-]} \int_{\Lambda/\zeta < k < \Lambda} \mathcal{D}\phi^+ e^{-F_0[\phi^+]} (1 + \lambda_0 \cos(\beta_0(\phi^- + \phi^+)) + \mathcal{O}(\lambda_0^2)). \quad (4.5)$$

Therefore, we find

$$\begin{aligned} F'[\phi^-] &= \int d^2x \left[\frac{1}{2}(\nabla\phi^-)^2 - \log \left(1 + \lambda_0 \langle \cos(\beta_0(\phi^- + \phi^+)) \rangle_+ + \mathcal{O}(\lambda_0^2) \right) \right] \\ &= \int d^2x \left[\frac{1}{2}(\nabla\phi^-)^2 - \lambda_0 \langle \cos(\beta_0(\phi^- + \phi^+)) \rangle_+ + \mathcal{O}(\lambda_0^2) \right]. \end{aligned} \quad (4.6)$$

The correlation function can be written as

$$\langle \cos(\beta_0(\phi^- + \phi^+)) \rangle_+ = \frac{1}{2} \left[\langle e^{i\beta_0\phi^+} \rangle_+ e^{i\beta_0\phi^-} + \langle e^{-i\beta_0\phi^+} \rangle_+ e^{-i\beta_0\phi^-} \right]. \quad (4.7)$$

Using the Wick identity, we find

$$\langle e^{i\beta_0\phi^+} \rangle_+ = \langle e^{-i\beta_0\phi^+} \rangle_+ = \exp \left(-\frac{1}{2}\beta_0^2 \langle \phi^+(0)\phi^+(0) \rangle_+ \right) = \exp \left(-\frac{1}{4\pi}\beta_0^2 \log \zeta \right). \quad (4.8)$$

Plugging back into the free energy, we find

$$F'[\phi^-] = \int d^2x \left[\frac{1}{2}(\nabla\phi^-)^2 - \lambda_0 \zeta^{-\beta_0^2/4\pi} \cos(\beta_0\phi^-) \right]. \quad (4.9)$$

Rescaling the UV cutoff back to Λ , namely $\mathbf{k}' = \zeta \mathbf{k}$ and $\mathbf{x}' = \mathbf{x}/\zeta$, we find

$$\lambda(\zeta) = \zeta^{-\beta_0^2/4\pi+2} \lambda_0. \quad (4.10)$$

Therefore, the potential is relevant when $\beta_0^2 < 8\pi$ and irrelevant when $\beta_0^2 > 8\pi$.

5. Sheet 3, Problem 5: Renormalisation of a Fluctuating Membrane

For a two-dimensional membrane in three-dimensional space, use

$$F[h] = \int d^2x \left[\frac{r_0}{2}(\nabla h)^2 + \frac{\lambda_0}{2}(1 + (\nabla h)^2)^{-5/2}(\nabla^2 h)^2 \right]. \quad (5.1)$$

Expand it through quartic order as

$$F[h] \simeq \int d^2x \left[\frac{r_0}{2}(\nabla h)^2 + \frac{\lambda_0}{2}(\nabla^2 h)^2 - \frac{5\lambda_0}{4}(\nabla h)^2(\nabla^2 h)^2 + \dots \right]. \quad (5.2)$$

Find the engineering dimensions of h, r_0, λ_0 . Separate $F = F_0 + F_I$ into quadratic and quartic pieces and split $h_k = h_k^- + h_k^+$ with $\Lambda/\zeta < |k| < \Lambda$ for h_k^+ . Derive the Gaussian fast-mode covariance

$$\langle h_{k_1}^+ h_{k_2}^+ \rangle_+ = \frac{(2\pi)^2 \delta(k_1 + k_2)}{r_0 k_1^2 + \lambda_0 k_1^4}. \quad (5.3)$$

Fourier transform F_I , isolate the term that renormalises $(\nabla^2 h)^2$ while ignoring field renormalisation, and derive $d\lambda/ds \simeq -5/(4\pi)$ for $\zeta = e^s$. Does short-distance fluctuation make the membrane more or less flexible in the infrared?

Solution. The engineering dimensions are $[h] = 1$, $[r_0] = -2$, and $[\lambda_0] = 0$. We read off $F_0[h_{\mathbf{k}}^+]$, which is

$$F_0[h_{\mathbf{k}}^+] = \int \frac{d^2k}{(2\pi)^2} \left(\frac{r_0}{2} k^2 + \frac{\lambda_0}{2} k^4 \right) h_{\mathbf{k}}^+ h_{-\mathbf{k}}^+. \quad (5.4)$$

This immediately gives

$$\langle h_{k_1}^+ h_{k_2}^+ \rangle_+ = \frac{(2\pi)^2 \delta(k_1 + k_2)}{r_0 k_1^2 + \lambda_0 k_1^4}. \quad (5.5)$$

The interaction free energy is

$$F_I[h_{\mathbf{k}}] = \frac{5\lambda_0}{4} \int \prod_{i=1}^4 \frac{d^2 k_i}{(2\pi)^2} \mathbf{k}_1 \cdot \mathbf{k}_2 k_3^2 k_4^2 h_{\mathbf{k}_1} h_{\mathbf{k}_2} h_{\mathbf{k}_3} h_{\mathbf{k}_4} (2\pi)^2 \delta^2 \left(\sum_i \mathbf{k}_i \right). \quad (5.6)$$

We shall focus only on the lowest order here. Using

$$F'[h_{\mathbf{k}}^-] = F_0[h_{\mathbf{k}}^-] + \langle F_I[h_{\mathbf{k}}^+, h_{\mathbf{k}}^-] \rangle_+ + \dots, \quad (5.7)$$

the only contributing term in $\langle F_I \rangle$ at this order is

$$\frac{5\lambda_0}{4} \int \prod_{i=1}^4 \frac{d^2 k_i}{(2\pi)^2} \mathbf{k}_1 \cdot \mathbf{k}_2 k_3^2 k_4^2 \langle h_{\mathbf{k}_1}^+ h_{\mathbf{k}_2}^+ \rangle_+ h_{\mathbf{k}_3}^- h_{\mathbf{k}_4}^- (2\pi)^2 \delta^2 \left(\sum_i \mathbf{k}_i \right). \quad (5.8)$$

Note that there is no polynomial factors here. This gives

$$-\frac{5\lambda_0}{4} \int \frac{d^2 k}{(2\pi)^2} \frac{d^2 q}{(2\pi)^2} k^4 \frac{1}{r_0 + \lambda_0 q^2} h_{\mathbf{k}}^- h_{-\mathbf{k}}^-. \quad (5.9)$$

Since λ_0 has engineering dimension zero, and the canonical term remains unchanged, we find

$$\lambda(\zeta) = \lambda_0 - 2 \cdot \frac{5\lambda_0}{4(2\pi)} \int_{\Lambda/\zeta}^{\Lambda} dq \frac{q}{r_0 + \lambda_0 q^2} + \dots. \quad (5.10)$$

Recall

$$\frac{d}{ds} \int_{\Lambda e^{-s}}^{\Lambda} dq f(q) = \Lambda f(\Lambda), \quad (5.11)$$

which gives

$$\frac{d\lambda}{ds} = -\frac{5\lambda_0}{4\pi} \frac{\Lambda^2}{r_0 + \lambda_0 \Lambda^2} + \dots \approx -\frac{5}{4\pi}, \quad (5.12)$$

where we have assumed $\Lambda^2 \gg r_0/\lambda_0$. Therefore, by integrating out the short distance behaviour, λ_0 , which measures the stiffness of the membrane, becomes smaller. That is, short-distance fluctuation makes the membrane more flexible in the IR.

6. Sheet 3, Problem 6: Integrating Out Two Coupled Order Parameters

Consider fields ϕ_1, ϕ_2 with independent sign flips and

$$F[\phi_1, \phi_2] = \int d^d x \left[\sum_{i=1}^2 \left(\frac{1}{2} (\nabla \phi_i)^2 + \frac{\mu_i^2}{2} \phi_i^2 + g_i \phi_i^4 \right) + \lambda \phi_1^2 \phi_2^2 \right]. \quad (6.1)$$

Set $\mu_i^2 = 0$ and integrate out short-wavelength modes in $d = 4 - \epsilon$. Show, for a common constant I , that

$$\frac{dg_i}{ds} = \epsilon g_i - (36g_i^2 + \lambda^2)I, \quad \frac{d\lambda}{ds} = \epsilon \lambda - (8\lambda^2 + 12\lambda g_1 + 12\lambda g_2)I. \quad (6.2)$$

Find the fixed points and show that the stable one has enhanced $O(2)$ symmetry.

Solution. Recall

$$F'[\phi_i^-] = F[\phi_i^-] + \langle F_I \rangle_+ - \frac{1}{2} [\langle F_I^2 \rangle_+ - \langle F_I \rangle_+^2] + \dots. \quad (6.3)$$

We shall keep $\mu_i^2 = 0$ through out the calculation, and hence neglect the running of it as well. $\langle F_I \rangle_+$ has no contributing terms to the running of λ, g_i . The leading order contribution comes from $\frac{1}{2} [\langle F_I^2 \rangle_+ - \langle F_I \rangle_+^2]$. The contributing diagrams are

$$\begin{aligned}
\delta g_1 : & \quad \underbrace{\text{diagram 1}}_{36g_1^2} + \underbrace{\text{diagram 2}}_{\lambda^2} \\
\delta g_2 : & \quad \underbrace{\text{diagram 3}}_{36g_2^2} + \underbrace{\text{diagram 4}}_{\lambda^2} \\
\delta \lambda : & \quad \underbrace{\text{diagram 5}}_{12\lambda g_1} + \underbrace{\text{diagram 6}}_{12\lambda g_2} + \underbrace{\text{diagram 7}}_{8\lambda^2},
\end{aligned}$$

where the black line denotes ϕ_1 and the red line denotes ϕ_2 . The factor of 36, for example, comes from $\binom{4}{2}^2$. All of these diagrams involves the integral

$$f(\mathbf{k}_1 + \mathbf{k}_2) = \int_{\Lambda/\zeta}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{k^2(\mathbf{k} + \mathbf{k}_1 + \mathbf{k}_2)^2} = \int_{\Lambda/\zeta}^{\Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{k^4} (1 + \mathcal{O}(\mathbf{k}_1, \mathbf{k}_2)), \quad (6.4)$$

where we can ignore terms including external momenta, since these generate terms of the form $(\phi^-)^2 \nabla^2 (\phi^-)^2$. These are irrelevant terms which we are not interested in. Since the couplings has engineering dimensions $[g_i] = [\lambda] = -\epsilon$, by rescaling we find

$$g_i(\zeta) = \zeta^\epsilon [g_i - (36g_i^2 + \lambda^2)f(0)] + \dots, \quad (6.5)$$

$$\lambda(\zeta) = \zeta^\epsilon [\lambda - (12\lambda g_1 + 12\lambda g_2 + 8\lambda^2)f(0)] + \dots. \quad (6.6)$$

Using

$$\frac{df(0)}{ds} = \frac{\Omega_{d-1}}{(2\pi)^d} \Lambda^{d-4} = \frac{\Lambda^{-\epsilon}}{8\pi^2}. \quad (6.7)$$

Introducing the dimensionless couplings $\tilde{g}_i = \Lambda^{-\epsilon} g_i$ and $\tilde{\lambda} = \Lambda^{-\epsilon} \lambda$ and differentiating with respect to s , we find

$$\frac{dg_i}{ds} = \epsilon g_i - \frac{1}{8\pi^2} (36g_i^2 + \lambda^2), \quad \frac{d\lambda}{ds} = \epsilon \lambda - \frac{1}{8\pi^2} (8\lambda^2 + 12\lambda g_1 + 12\lambda g_2), \quad (6.8)$$

where we have suppressed the tilde.

Let's find the fixed points of the theory by solving

$$0 = \epsilon g_1 - \frac{1}{8\pi^2} (36g_1^2 + \lambda^2), \quad (6.9)$$

$$0 = \epsilon g_2 - \frac{1}{8\pi^2} (36g_2^2 + \lambda^2), \quad (6.10)$$

$$0 = \lambda \left(\epsilon - \frac{1}{8\pi^2} (8\lambda + 12g_1 + 12g_2) \right). \quad (6.11)$$

Clearly, it has a trivial Gaussian fixed point $\lambda = g_i = 0$.

For $\lambda = 0$, we have

$$g_i = 0 \quad \text{or} \quad g_i = \frac{2\pi^2}{9} \epsilon, \quad (6.12)$$

giving the three additional fixed points

$$(g_1, g_2, \lambda) = \left(\frac{2\pi^2}{9} \epsilon, 0, 0 \right), \quad \left(0, \frac{2\pi^2}{9} \epsilon, 0 \right), \quad \left(\frac{2\pi^2}{9} \epsilon, \frac{2\pi^2}{9} \epsilon, 0 \right). \quad (6.13)$$

For $\lambda \neq 0$, combining the first two equations gives

$$(g_1 - g_2) \left[\epsilon - \frac{9}{2\pi^2}(g_1 + g_2) \right] = 0. \quad (6.14)$$

The nontrivial solutions have $g_1 = g_2 = g$. The third equation then gives

$$\lambda = \pi^2 \epsilon - 3g. \quad (6.15)$$

Substituting this into the first equation gives

$$45g^2 - 14\pi^2 \epsilon g + \pi^4 \epsilon^2 = 0, \quad (6.16)$$

and hence

$$g_1 = g_2 = \frac{\pi^2}{5} \epsilon, \quad \lambda = \frac{2\pi^2}{5} \epsilon, \quad (6.17)$$

or

$$g_1 = g_2 = \frac{\pi^2}{9} \epsilon, \quad \lambda = \frac{2\pi^2}{3} \epsilon. \quad (6.18)$$

Expanding about

$$\lambda = \lambda_* + \delta\lambda, \quad g_i = g_{i*} + \delta g_i,$$

the linearised flow is

$$\frac{d}{ds} \begin{pmatrix} \delta\lambda \\ \delta g_1 \\ \delta g_2 \end{pmatrix} = M \begin{pmatrix} \delta\lambda \\ \delta g_1 \\ \delta g_2 \end{pmatrix}. \quad (6.19)$$

The stability matrix is

$$M = \begin{pmatrix} \epsilon - \frac{4\lambda_* + 3g_{1*} + 3g_{2*}}{2\pi^2} & -\frac{3\lambda_*}{2\pi^2} & -\frac{3\lambda_*}{2\pi^2} \\ -\frac{\lambda_*}{4\pi^2} & \epsilon - \frac{9g_{1*}}{\pi^2} & 0 \\ -\frac{\lambda_*}{4\pi^2} & 0 & \epsilon - \frac{9g_{2*}}{\pi^2} \end{pmatrix}.$$

At the Gaussian fixed point,

$$M_G = \epsilon \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad (6.20)$$

so all three eigenvalues are positive. Hence the Gaussian fixed point is unstable in all directions. For the fixed point

$$(g_{1*}, g_{2*}, \lambda_*) = \left(\frac{2\pi^2}{9} \epsilon, 0, 0 \right), \quad (6.21)$$

we find

$$M = \epsilon \begin{pmatrix} \frac{2}{3} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6.22)$$

The eigenvalues and corresponding eigenvectors are

$$\omega_1 = \frac{2}{3}\epsilon, \quad v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \omega_2 = -\epsilon, \quad v_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \omega_3 = \epsilon, \quad v_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (6.23)$$

Thus this fixed point has one stable and two unstable directions. The fixed point

$$(g_{1*}, g_{2*}, \lambda_*) = \left(0, \frac{2\pi^2}{9} \epsilon, 0 \right) \quad (6.24)$$

is equivalent under $g_1 \leftrightarrow g_2$ and has the same stability properties.

For the decoupled fixed point

$$(g_{1*}, g_{2*}, \lambda_*) = \left(\frac{2\pi^2}{9}\epsilon, \frac{2\pi^2}{9}\epsilon, 0 \right), \quad (6.25)$$

we obtain

$$M_{\text{dec}} = \epsilon \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \quad (6.26)$$

Hence

$$\omega_1 = \frac{1}{3}\epsilon, \quad \omega_2 = \omega_3 = -\epsilon, \quad (6.27)$$

with eigenvectors along the λ , g_1 , and g_2 directions respectively. Therefore, this fixed point is stable in the g_1 and g_2 directions but unstable to switching on λ .

For

$$g_{1*} = g_{2*} = \frac{\pi^2}{9}\epsilon, \quad \lambda_* = \frac{2\pi^2}{3}\epsilon, \quad (6.28)$$

we find

$$M = \epsilon \begin{pmatrix} -\frac{2}{3} & -1 & -1 \\ -\frac{1}{6} & 0 & 0 \\ -\frac{1}{6} & 0 & 0 \end{pmatrix}. \quad (6.29)$$

Its eigenvalues are

$$\omega_1 = -\epsilon, \quad \omega_2 = 0, \quad \omega_3 = \frac{1}{3}\epsilon, \quad (6.30)$$

with corresponding eigenvectors

$$v_1 = \begin{pmatrix} 6 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}. \quad (6.31)$$

This fixed point therefore has a stable direction, an unstable direction, and a marginal direction at this order, and is not IR stable.

Finally, consider

$$g_{1*} = g_{2*} = \frac{\pi^2}{5}\epsilon, \quad \lambda_* = \frac{2\pi^2}{5}\epsilon. \quad (6.32)$$

The stability matrix is

$$M_{O(2)} = \epsilon \begin{pmatrix} -\frac{2}{5} & -\frac{3}{5} & -\frac{3}{5} \\ -\frac{1}{10} & -\frac{4}{5} & 0 \\ -\frac{1}{10} & 0 & -\frac{4}{5} \end{pmatrix}. \quad (6.33)$$

Its eigenvalues are

$$\omega_1 = -\epsilon, \quad \omega_2 = -\frac{4}{5}\epsilon, \quad \omega_3 = -\frac{1}{5}\epsilon \quad (6.34)$$

with corresponding eigenvectors

$$v_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -6 \\ 1 \\ 1 \end{pmatrix}. \quad (6.35)$$

Since all three eigenvalues are negative, this fixed point is IR stable.

Moreover, at this fixed point $\lambda_* = 2g_*$, so the quartic interaction becomes

$$g_*(\phi_1^4 + \phi_2^4) + \lambda_*\phi_1^2\phi_2^2 = g_*(\phi_1^2 + \phi_2^2)^2. \quad (6.36)$$

Thus the stable fixed point has an enhanced $O(2)$ symmetry, as required.